VMware Workstation 17

Lab Environment Setup Guide

Domain Controller + SQL Server 2019 Multi-Node Configuration

Host Machine	PMSQLDBA
Network	192.168.0.0/24 (Bridged)
Hypervisor	VMware Workstation 17
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1. Lab Architecture Overview

This guide provides a complete end-to-end configuration walkthrough for setting up a Windows Server Active Directory Domain Controller and three SQL Server 2019 nodes inside VMware Workstation 17, using a Bridged network that allows all VMs to communicate with each other and with your physical host machine (PMSQLDBA).

1.1 VM Inventory

The following four virtual machines make up the lab environment:

VM Name	OS / Role	vCPU	RAM	Disk	Notes
DC	Windows Server 2022 / AD DS	2	4 GB	60 GB	Domain Controller + DNS
VMSQL01	Windows Server 2022 + SQL 2019	4	8 GB	100 GB	Primary SQL Node
VMSQL02	Windows Server 2022 + SQL 2019	4	8 GB	100 GB	Secondary SQL Node
VMSQL03	Windows Server 2022 + SQL 2019	4	8 GB	100 GB	Tertiary SQL Node

1.2 IP Address Plan

All VMs use static IPs within your existing home/office subnet (192.168.0.x). Reserve these in your router's DHCP exclusion list to prevent conflicts.

VM Name	Role	IP Address	Subnet Mask	Gateway	DNS
DC	Domain Controller / DNS	192.168.0.10	255.255.255.0	192.168.0.1	127.0.0.1
VMSQL01	SQL Server Node 1	192.168.0.11	255.255.255.0	192.168.0.1	192.168.0.10
VMSQL02	SQL Server Node 2	192.168.0.12	255.255.255.0	192.168.0.1	192.168.0.10
VMSQL03	SQL Server Node 3	192.168.0.13	255.255.255.0	192.168.0.1	192.168.0.10

> ⚠ **WARNING** Reserve 192.168.0.10 – 192.168.0.13 in your router's DHCP exclusion range to avoid IP conflicts with other DHCP-assigned devices on the same network.

2. VMware Workstation – Bridged Network Configuration

Bridged networking connects your VMs directly to your physical network through the host's Wi-Fi adapter (Intel Wi-Fi 6 AX201), making them appear as independent physical machines on the 192.168.0.0/24 subnet.

2.1 Configure Virtual Network Editor

Step 1: Open VMware Workstation 17.

Step 2: Go to Edit > Virtual Network Editor.

Step 3: Click Change Settings (requires Administrator privileges).

Step 4: Select VMnet0 (Bridged) from the list.

Step 5: Under Bridged to, select Intel(R) Wi-Fi 6 AX201 160MHz from the dropdown.

Step 6: Click Apply, then OK.

[SCREENSHOT PLACEHOLDER]
VMware Virtual Network Editor – VMnet0 bridged to Intel Wi-Fi 6 AX201

NOTE If you see Automatic in the Bridged to field, change it to explicitly select your Wi-Fi adapter. Automatic can sometimes bridge to the wrong adapter (e.g., Fortinet VPN adapter which is disconnected).

2.2 Assign Bridged Adapter to Each VM

Repeat the following for DC, VMSQL01, VMSQL02, and VMSQL03:

Step 1: Right-click the VM > Settings.

Step 2: Select Network Adapter.

Step 3: Set Connection to Bridged (Automatic).

Step 4: Check Replicate physical network connection state.

Step 5: Click OK.

[SCREENSHOT PLACEHOLDER]
VM Settings – Network Adapter set to Bridged

3. Domain Controller (DC) Configuration

3.1 Set Static IP Address on DC

Step 1: Power on the DC VM and log in.

Step 2: Open Control Panel > Network and Internet > Network Connections.

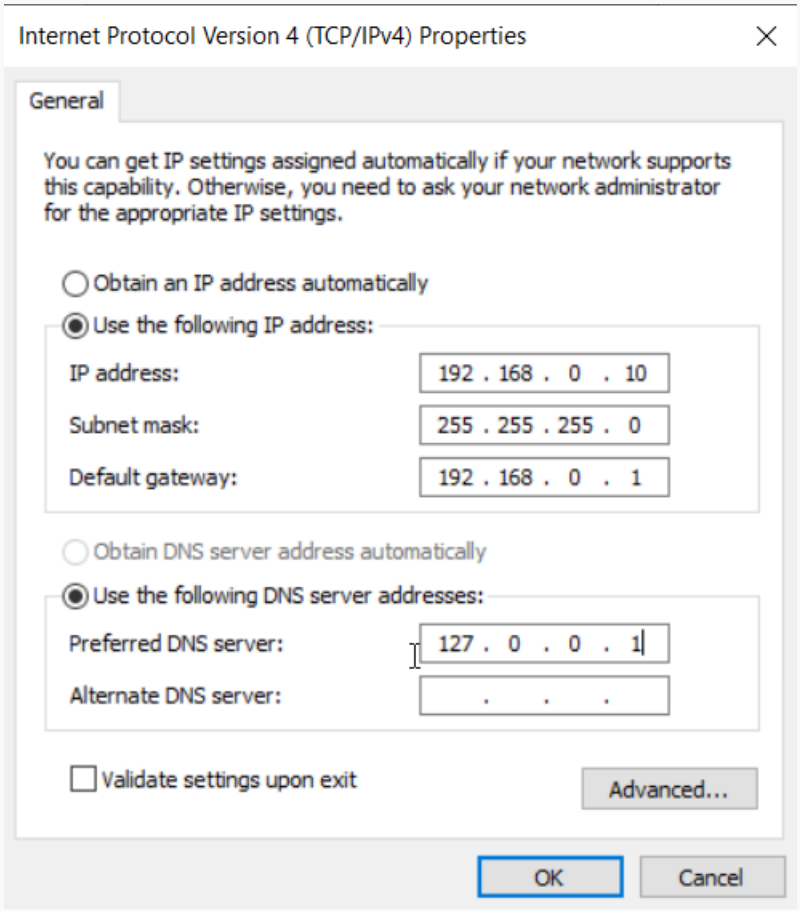
Step 3: Right-click Ethernet > Properties.

Step 4: Select Internet Protocol Version 4 (TCP/IPv4) > Properties.

Step 5: Enter the following values:

IP Address	192.168.0.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.0.1
Preferred DNS	127.0.0.1
Alternate DNS	(leave blank)

[SCREENSHOT PLACEHOLDER]
TCP/IPv4 Properties – Static IP configured on DC



Step 6: Click OK > Close to save.

3.2 Rename the Domain Controller

Step 1: Right-click Start > System.

Step 2: Click Rename this PC.

Step 3: Enter DC as the computer name.

Step 4: Click Next > Restart Now.

[SCREENSHOT PLACEHOLDER]
System Properties – Computer Name set to DC

3.3 Install Active Directory Domain Services (AD DS)

Step 1: Open Server Manager > Manage > Add Roles and Features.

Step 2: Click Next through the wizard until Server Roles.

Step 3: Check Active Directory Domain Services.

Step 4: Click Add Features when prompted.

Step 5: Continue clicking Next and then Install.

Step 6: Wait for installation to complete. Do NOT restart yet.

[SCREENSHOT PLACEHOLDER]
Add Roles and Features – Active Directory Domain Services selected

3.4 Promote Server to Domain Controller

Step 1: In Server Manager, click the flag notification icon > Promote this server to a domain controller.

Step 2: Select Add a new forest.

Step 3: Enter Root domain name: e.g., lab.local (or your preferred domain).

Step 4: Click Next.

[SCREENSHOT PLACEHOLDER]
AD DS Configuration Wizard – Add a new forest with domain name lab.local

Step 5: On Domain Controller Options, set Forest and Domain functional levels to Windows Server 2016.

Step 6: Check Domain Name System (DNS) server.

Step 7: Enter a strong DSRM password and confirm.

Step 8: Click Next through the DNS Options and Additional Options pages.

Step 9: On the Paths page, leave defaults and click Next.

Step 10: Review the summary and click Install.

Step 11: The server will restart automatically.

[SCREENSHOT PLACEHOLDER]
DC Options – DNS + DSRM password configuration

[SCREENSHOT PLACEHOLDER]
Prerequisites Check passed – Ready to Install

NOTE After the DC reboots, log in with DOMAIN\Administrator or lab\Administrator. Active Directory Users and Computers and DNS Manager will now be available in Server Manager > Tools.

3.5 Verify DNS and AD DS

Step 1: Open Server Manager > Tools > DNS Manager.

Step 2: Expand your server name > Forward Lookup Zones.

Step 3: Confirm lab.local (or your domain) appears.

Step 4: Open Active Directory Users and Computers.

Step 5: Confirm the lab.local domain tree is visible.

[SCREENSHOT PLACEHOLDER]
DNS Manager – Forward Lookup Zone lab.local created

4. SQL Server VMs – Static IP & Hostname Configuration

Repeat Sections 4.1 and 4.2 for VMSQL01, VMSQL02, and VMSQL03, substituting the respective IP addresses from the IP Address Plan in Section 1.2.

4.1 Set Static IP Address (per SQL VM)

Use the following IP addresses for each VM:

VM	IP Address	Subnet Mask	Gateway	Preferred DNS
VMSQL01	192.168.0.11	255.255.255.0	192.168.0.1	192.168.0.10
VMSQL02	192.168.0.12	255.255.255.0	192.168.0.1	192.168.0.10
VMSQL03	192.168.0.13	255.255.255.0	192.168.0.1	192.168.0.10

Step 1: Open Network Connections > Ethernet Properties > IPv4 Properties.

Step 2: Enter the static IP for that VM (see table above).

Step 3: Set Preferred DNS to 192.168.0.10 (the DC's IP).

Step 4: Click OK to save.

[SCREENSHOT PLACEHOLDER]
VMSQL01 – TCP/IPv4 Static IP 192.168.0.11 with DNS pointing to DC

4.2 Rename Each SQL VM

Step 1: Right-click Start > System > Rename this PC.

Step 2: Enter VMSQL01 (or VMSQL02 / VMSQL03 respectively).

Step 3: Click Next > Restart Now.

[SCREENSHOT PLACEHOLDER]
System Properties – Rename PC to VMSQL01

5. Joining SQL VMs to the Domain

With the DC promoted and DNS configured, join each SQL VM to the lab.local domain. Perform the following steps on VMSQL01, VMSQL02, and VMSQL03.

5.1 Verify DNS Resolution Before Joining

Step 1: Open Command Prompt on the SQL VM.

Step 2: Run: nslookup lab.local

Step 3: The result should return 192.168.0.10 (DC IP). If not, verify the DNS setting is correct.

Step 4: Run: ping 192.168.0.10 – confirm 4 replies with no loss.

[SCREENSHOT PLACEHOLDER]
CMD – nslookup lab.local returns 192.168.0.10

⚠ **WARNING** If nslookup fails, ensure the Preferred DNS on the SQL VM is set to 192.168.0.10 and that the DC is powered on and accessible.

5.2 Join the Domain

Step 1: Right-click Start > System.

Step 2: Scroll down to Advanced system settings or click Rename this PC (advanced).

Step 3: Click the Computer Name tab > Change.

Step 4: Under Member of, select Domain.

Step 5: Enter lab.local and click OK.

[SCREENSHOT PLACEHOLDER]
System Properties – Domain join dialog showing lab.local

Step 6: Enter credentials: Username: Administrator, Password: (DC administrator password).

Step 7: Click OK. You should see: Welcome to the lab.local domain.

Step 8: Click OK > Close > Restart Now.

[SCREENSHOT PLACEHOLDER]
Welcome to the lab.local domain – successful domain join message

> **NOTE** Repeat this section for VMSQL02 and VMSQL03. Each VM must be restarted individually after joining the domain.

5.3 Verify Domain Membership from DC

Step 1: On the DC, open Active Directory Users and Computers.

Step 2: Expand lab.local > Computers.

Step 3: Confirm VMSQL01, VMSQL02, and VMSQL03 appear in the list.

[SCREENSHOT PLACEHOLDER]
Active Directory Users and Computers – Computers OU showing VMSQL01, VMSQL02, VMSQL03

6. Connectivity Verification – Ping Tests

Before proceeding to SQL Server installation, verify full network connectivity between all VMs using ping from each machine.

6.1 Ping Matrix

Run the following pings from each VM and confirm all replies succeed (0% packet loss):

From VM	Ping Target	Expected IP	Expected Result
VMSQL01	ping DC	192.168.0.10	Reply from 192.168.0.10
VMSQL01	ping VMSQL02	192.168.0.12	Reply from 192.168.0.12
VMSQL01	ping VMSQL03	192.168.0.13	Reply from 192.168.0.13
VMSQL02	ping DC	192.168.0.10	Reply from 192.168.0.10
VMSQL02	ping VMSQL01	192.168.0.11	Reply from 192.168.0.11
VMSQL02	ping VMSQL03	192.168.0.13	Reply from 192.168.0.13
VMSQL03	ping DC	192.168.0.10	Reply from 192.168.0.10
VMSQL03	ping VMSQL01	192.168.0.11	Reply from 192.168.0.11
VMSQL03	ping VMSQL02	192.168.0.12	Reply from 192.168.0.12
DC	ping VMSQL01	192.168.0.11	Reply from 192.168.0.11
DC	ping VMSQL02	192.168.0.12	Reply from 192.168.0.12
DC	ping VMSQL03	192.168.0.13	Reply from 192.168.0.13

[SCREENSHOT PLACEHOLDER]
CMD – Ping test from VMSQL01 to DC, VMSQL02, VMSQL03 – all successful

⚠ **WARNING** If ping fails, check Windows Firewall on the target VM. Enable File and Printer Sharing (Echo Request – ICMPv4-In) rule via Windows Firewall with Advanced Security.

6.2 Enable ICMP Ping via Firewall (if blocked)

Step 1: On each VM, open Windows Defender Firewall with Advanced Security.

Step 2: Go to Inbound Rules.

Step 3: Find File and Printer Sharing (Echo Request – ICMPv4-In).

Step 4: Right-click > Enable Rule.

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Windows Firewall – ICMPv4 Echo Request Inbound Rule enabled

7. SQL Server 2019 Installation

Install SQL Server 2019 on VMSQL01, VMSQL02, and VMSQL03. Use a service account created in Active Directory for consistent and secure SQL Server service management.

7.1 Create SQL Service Account in Active Directory

On the DC, perform the following:

Step 1: Open Active Directory Users and Computers.

Step 2: Right-click Users > New > User.

Step 3: Enter: First name: SQL, Last name: Service, User logon name: svc_sql

Step 4: Click Next. Set a strong password.

Step 5: Check Password never expires. Uncheck User must change password at next logon.

Step 6: Click Finish.

[SCREENSHOT PLACEHOLDER]
ADUC – New User svc_sql created in lab.local\Users

7.2 Install SQL Server 2019 (per SQL VM)

Step 1: Mount the SQL Server 2019 ISO or insert the installation media.

Step 2: Run setup.exe. Click Installation > New SQL Server stand-alone installation.

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SQL Server 2019 Installation Center – New Stand-alone Installation

Step 3: Accept the license terms. Click Next.

Step 4: On Feature Selection, check:

- Database Engine Services
- SQL Server Replication
- Full-Text and Semantic Extractions for Search
- SQL Server Management Tools – Basic (if SSMS not installed separately)

[SCREENSHOT PLACEHOLDER]
SQL Setup – Feature Selection with Database Engine Services checked

Step 5: On Instance Configuration, select Default Instance (MSSQLSERVER) or name it.

Step 6: On Server Configuration > Service Accounts tab, set:

- SQL Server Agent: LAB\svc_sql
- SQL Server Database Engine: LAB\svc_sql

[SCREENSHOT PLACEHOLDER]
SQL Setup – Service Accounts tab with svc_sql domain account

Step 7: On Database Engine Configuration > Server Configuration tab:

- Authentication Mode: Mixed Mode
- Enter a strong SA password
- Add current user and LAB\Administrator as SQL Server Administrators

[SCREENSHOT PLACEHOLDER]
SQL Setup – Database Engine Configuration – Mixed Mode + Admin accounts

Step 8: Click Next > Install. Wait for installation to complete (10–20 minutes).

Step 9: Click Close when installation succeeds.

NOTE Repeat Section 7.2 for VMSQL02 and VMSQL03. Use the same svc_sql service account and SA password across all three nodes for consistency.

8. Post-Installation Verification

8.1 Verify SQL Server Service is Running

Step 1: On each SQL VM, open Services (services.msc).

Step 2: Confirm SQL Server (MSSQLSERVER) status is Running.

Step 3: Confirm SQL Server Agent status is Running.

[SCREENSHOT PLACEHOLDER]
Services.msc – SQL Server (MSSQLSERVER) and SQL Server Agent running

8.2 Connect via SSMS

Step 1: Open SQL Server Management Studio (SSMS).

Step 2: Server name: 192.168.0.11 (or VMSQL01\MSSQLSERVER).

Step 3: Authentication: SQL Server Authentication, Login: sa.

Step 4: Click Connect. Confirm Object Explorer shows the server.

[SCREENSHOT PLACEHOLDER]
SSMS – Connected to VMSQL01 showing Object Explorer

NOTE For VMSQL02 and VMSQL03, use their respective IPs (192.168.0.12 and 192.168.0.13) to connect via SSMS.

8.3 Enable SQL Server Remote Connections & TCP/IP

Step 1: Open SQL Server Configuration Manager.

Step 2: Expand SQL Server Network Configuration > Protocols for MSSQLSERVER.

Step 3: Right-click TCP/IP > Enable.

Step 4: Restart the SQL Server service.

[SCREENSHOT PLACEHOLDER]
SQL Server Configuration Manager – TCP/IP Protocol Enabled

9. Summary Checklist

Use this checklist to confirm all steps have been completed successfully:

#	Task	Status
1	VMware Virtual Network Editor – VMnet0 bridged to Wi-Fi adapter	☐ Complete
2	All 4 VMs set to Bridged network adapter	☐ Complete
3	DC: Static IP 192.168.0.10 configured	☐ Complete
4	DC: Renamed to DC and restarted	☐ Complete
5	DC: AD DS role installed	☐ Complete
6	DC: Promoted to Domain Controller – lab.local	☐ Complete
7	DC: DNS Manager shows lab.local forward zone	☐ Complete
8	VMSQL01: Static IP 192.168.0.11, DNS to 192.168.0.10	☐ Complete
9	VMSQL02: Static IP 192.168.0.12, DNS to 192.168.0.10	☐ Complete
10	VMSQL03: Static IP 192.168.0.13, DNS to 192.168.0.10	☐ Complete
11	VMSQL01: Joined to lab.local domain	☐ Complete
12	VMSQL02: Joined to lab.local domain	☐ Complete
13	VMSQL03: Joined to lab.local domain	☐ Complete
14	All VMs appear in ADUC > Computers	☐ Complete
15	Ping tests pass between all 4 VMs	☐ Complete
16	SQL Server 2019 installed on VMSQL01	☐ Complete
17	SQL Server 2019 installed on VMSQL02	☐ Complete
18	SQL Server 2019 installed on VMSQL03	☐ Complete
19	svc_sql service account used on all SQL nodes	☐ Complete
20	TCP/IP protocol enabled on all SQL instances	☐ Complete
21	SSMS connects to all 3 SQL nodes successfully	☐ Complete

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